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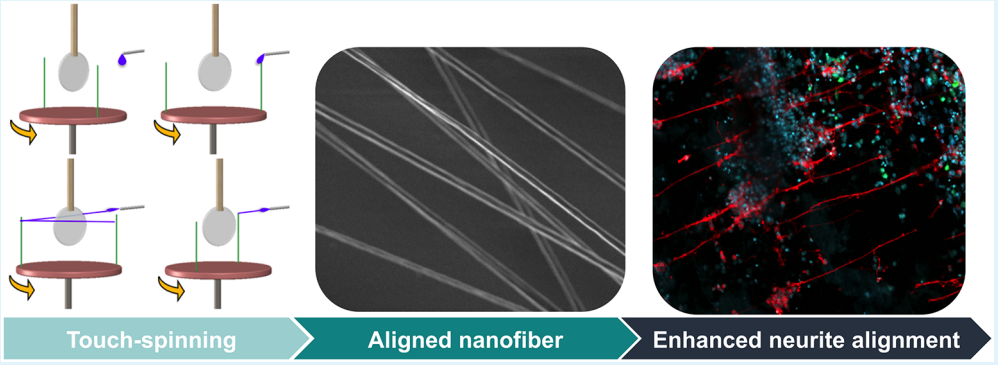
Touch-Spun Nanofibers for Nerve Regeneration

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ABSTRACT: In the current study, we examined the potential for neural stem cell (NSCs) proliferation on novel aligned touch-spun polycaprolactone (PCL) nanofibers. Electrospun PCL nanofibers with similar diameter and alignment were used as a control. Confocal microscopy images showed that NSCs grew and differentiated all over the scaffolds up to 8 days. Neurite quantification analysis revealed that the NSCs cultured on the touch-spun fibers with incorporated bovine serum albumin promoted the expression of neuron-specific class III β-tubulin after 8 days. More importantly, NSCs grown on the aligned touch-spun PCL fibers exhibited a bipolar elongation along the direction of the fiber, while NSCs cultured on the aligned electrospun PCL fibers expressed a multipolar elongation. The structural characteristics of the PCL nanofibers analyzed by X-ray diffraction indicated that the degree of crystallinity and elastic modulus of the touch-spun fiber are significantly higher than those of electrospun fibers. These findings indicate that the aligned and stiff touch-spun nanofibrous scaffolds show considerable potential for nerve injury repair.

KEYWORDS: *nanofiber, tissue engineering, nerve regeneration, touch-spinning, crystallinity, neurite extension*

1. INTRODUCTION

Fabrication of nanostructured scaffolds for biological applications has been very promising, as their nanoscale dimension mimics the natural extracellular matrix (ECM) where cells can interact efficiently.[1−5] The polymeric nanofibrous scaffolds are advantageous in regards to enhanced cell adhesion due to their high surface area and porosity.[6,7] Additionally, aligned fibers have provided better contact guidance effects on the neurite outgrowth of primary cortical neurons and PC-12 cells, which makes them effective scaffolds for spinal cord and peripheral nerve regeneration.[8−11] Despite the broad use of electrospinning technologies for the fabrication of nanofibrous scaffolds in the field of regenerative medicine, several reports have shown detrimental effects of the high voltage on the loaded biomolecules in the electrospun fibers, and as such, there have been challenges to retain the bioactivity of loaded drugs and biomolecules, such as bovine serum albumin (BSA) and heparin.[12−14] Hence, alternative techniques should be developed for the fabrication of nanofibrous scaffolds to induce comprehensive guidance of cells along the tissue-engineered grafts.

The touch-spinning technique has been developed recently to create polymer nanofibers without using high-voltage electric fields.[15] In this simple and controllable process, the polymer solution is pushed through a syringe and adhered to a rotating rod on a spinneret in order to draw the fiber from a droplet of the polymer solution. The fabricated fiber diameter is controlled precisely in the range of 40 nm to greater than 5 μm by adjusting the rotational speed, polymer concentration, and flow rate. The spinning setup can be used to wind

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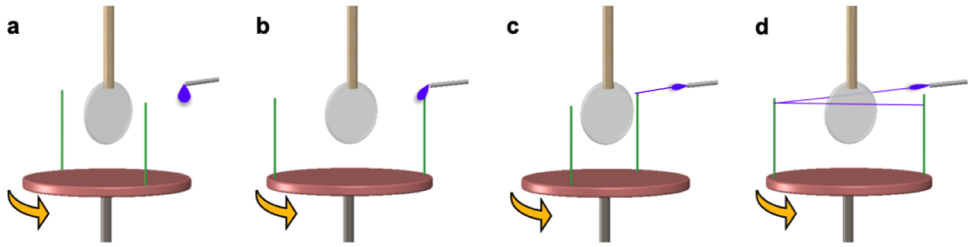


Figure 1. Touch-spinning setup: (a) the disk rotates with a controllable spinning rate and the rods on the disk are in close proximity to the polymer droplet, (b) when the disk rotates, the tip of the rod touches the polymer droplet, (c) the continuous mechanical force between the bar and the polymer droplet stretches the polymer droplet and a liquid bridge forms, and (d) the solvent evaporates and the fiber gets collected around the collector held in the center.

unidirectionally, orthogonally, and randomly oriented two-dimensional- and three-dimensional (3D)-meshes with controlled fiber density. Thus, due to its ease of use and adjustability, this fabrication method can be developed for the large-scale industrial manufacturing process.

Among a number of polymers explored for neural tissue engineering, polycaprolactone (PCL) is a well-known biocompatible polymer. Modifications of PCL by blending with other chemicals or growth factor proteins can closely mimic the physiological microenvironment, and these can enhance cell growth and differentiation.[16−20] As an example, a study by Tabernero et al. demonstrated that albumin can maintain long-term survival of neurons by increasing the synthesis and release of glutamate.[21] Here, we used BSA blended in a PCL solution to stimulate cell growth. Specifically, we examined the effects of BSA in both electrospun and touch-spun fibers on the neural cell growth and differential potential of neural stem cells (NSCs).

The degree of crystallinity is an important factor for polymer materials interacting with cells. Crystallinity affects density, elastic modulus, and other mechanical characteristics at the microscopic scale, which may influence cell contact guidance. There have been several reports of cellular response to polymers with different crystallinity properties,[22−25] but to the best of our knowledge, the influence of different crystallinity of PCL nanofibers on neural cell behavior has yet to be determined. Some studies have indicated that the electrospun fibers from polymer solutions exhibit a lower degree of crystallinity properties than fibers produced from melts.[26−28] The difference is likely due to the polymer chain alignment achieved during the fiber drawing stages. In this study, we investigated how the touch-spinning nanofiber fabrication method affects the degree of crystallinity of PCL nanofibers compared to electrospinning.

Mouse neuroepithelial cells (NE-4C) were selected for the cell model in the current study considering their key characteristic markers of neural stem cells (NSCs) such as nestin and tubulin β III immunoreactivities during the early stage of neural differentiation.[29,30] Retinoic acid (RA) promotes NSC lineage differentiation into different types of neurons and glial cells in a progressive process, in which they display specific morphological and physiological features.[29] However, to our knowledge, there have been no studies that have demonstrated the alignment of NE-4C cells at early neural stages.

This work demonstrates aligned touch-spun PCL nanofibers with a high degree of crystallinity to fabricate neural constructs, and it subsequently investigates the NSC growth and differentiation therein. Present study aims to (1) demonstrate the feasibility of utilizing a touch-spinning setup to create aligned nanofibrous neural guides, (2) to examine guided neurite extension along the aligned fiber, and to (3) evaluate and compare the effects of BSA in both electrospun and touch-spun fibers on the growth and differential potential of NSC in vitro.

2. EXPERIMENTAL SECTION

2.1. Polymer Solution Preparation. PCL (M_n = 80 000 g mol^{-1}) (Sigma-Aldrich) was dissolved in chloroform (ACS grade, BDH, VWR) for 1 h at 60 °C to produce a solution with a concentration of 8 wt %. The 0.01% BSA in the PCL solution was prepared by dispersing 0.1 mg of BSA (Sigma-Aldrich, lyophilized powder with MW ∼ 66 kDa) in 1 mL of chloroform at room temperature. Then, PCL (M_n = 80 000 g mol^{-1}) was added to the BSA solution in chloroform to prepare an 8 wt % solution.

2.2. Touch-Spun and Electrospun Nanofiber Preparation. For the touch-spun setup as described elsewhere,[31] a rotation speed of the spinneret was set at 1400 RPM, and the polymer solution flow rate was 10 μL min^{-1}. The fibers were fabricated at ambient temperature, and these were collected around a 12 mm glass circular coverslip (Fisher Scientific). Figure 1 shows a schematic diagram of our touch-spinning fabrication process.

Electrospun fiber samples were prepared by using an electrospinning setup (Spraybase) with a rotating drum collector (Profector Life Sciences Ltd.) with a rotation speed of 1400 RPM. An 8 wt % PCL solution without or with BSA was supplied at a flow speed of 10 μL min^{-1}. The needle (0.9 mm in diameter) was linked to a high voltage (20 kV), and a working distance was adjusted to 110 mm. The fibers were spun at room temperature, and they were collected on 12 mm glass circular coverslips.

2.3. Fiber Characterization. Prepared touch-spun and electrospun fibers were imaged under an FEI Teneo scanning electron microscope (SEM). A sputter coater (Cressington 208HR) was employed to sputter-coat all samples with iridium for 30 s at a coating rate of 0.1 nm s^{-1} prior to imaging. Subsequently, the iridium-coated samples were imaged by SEM at 5 kV with a current of 0.2 nA.

The crystal structures of the fabricated PCL fibers were characterized by an X-ray diffractometer (XRD) (PANalytical X'Pert PRO MRD). XRD measurements were performed at ambient temperature. An X-ray beam generated by a Genix3D-integrated microfocus fixed anode (Cu) source operated at 30 kV and 0.28 mA. Two sets of scatter-less slits were used to collimate the beam to a size of 8 × 8 mm at the sample. Scattering intensities were registered by a mar345 image plate detector at 20 cm distance from the sample with 1 h of X-ray exposure time. The data were analyzed using home-written Matlab codes. To estimate the crystallite size of the PCL fibers, the Scherrer equation $\langle L\rangle_{hkl} = \mathbf{K}\lambda/\beta\cos(\theta)$ is used, where $\langle L\rangle_{110}$ is the crystallite size estimated from the (hkl) reflection planes, K is the constant (0.9), λ is the wavelength of the X-ray source, β is the full width at half maximum (FWHM) peak intensity in radians, and θ is the Bragg angle (i.e., half of the scattering angle measured from the incident beam). To estimate FWHM for PCL fibers, the first diffraction peaks corresponding to (110) diffraction are fitted using a

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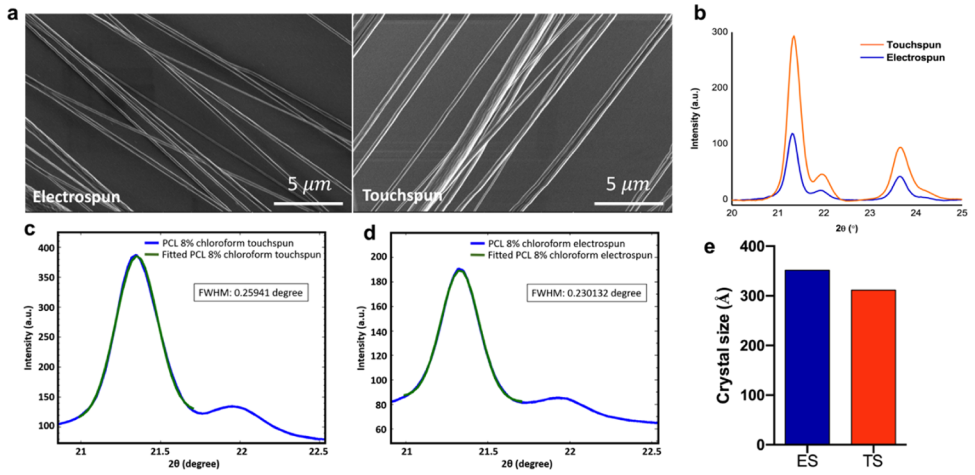


Figure 2. Fabricated nanofibers and characterization. (a) SEM images of aligned PCL nanofibers prepared by electrospinning and touch-spinning. (b) XRD pattern describing the crystallinity of PCL fibers prepared by touch-spinning and electrospinning. Scattering intensity $I(Q)$ as a function of scattering angle 2θ. The Gaussian fit of (110) peak of (c) touch-spun PCL fiber and of (d) electrospun PCL fibers. (e) Crystallite size of electrospun and touch-spun fibers. ES = electrospun fibers; TS = touch-spun fibers.

Table 1. Concentration of Elements Detected (in Atom %) for Electrospun (E) and Touch-Spun (T) Fibers vs Reference BSA and PCL Samples

content of BSA in the PCL solution %	fabrication method	CH_x	C−(O,N)	(N,O)−C═O	C_{total}	N	O	minor elements
5	E	46.1	13.7	10.7	70.6	3.2	26.2	P, Cl, Na
5	T	51.1	13.0	10.8	74.8	0.3	24.9	Si, Cl, Na
0.1	E	45.0	15.8	11.7	72.5	3.2	26.2	P, Si, Cl, Na
0.1	T	55.8	12.3	10.5	72.1	1.4	23.1	Na, Cl
100% BSA	reference	38.6	18.2	13.9	70.7	12.6	16.6	
100% PCL	reference	50.0	12.5	12.5	75.0		25.0	

Gaussian function. Figure 2b−d shows the fitted (110) diffraction peaks with FWHM in degrees, and Table 1 shows the crystallite size of the PCL fibers.

X-ray photoelectron spectroscopy (XPS) measurements were carried out using a Physical Electronics VeraProbe II instrument equipped with a monochromatic Al kα X-ray source ($h\nu$ = 1486.7 eV) and a concentric hemispherical analyzer. PCL fibers without and with BSA were immersed in phosphate-buffered saline (PBS) for 8 days and air-dried for XPS analysis. Both low-energy electrons (<5 eV) and argon ions were used to conduct charge neutralization. Prior to the measurement, calibration was performed on the binding energy axis using sputter-cleaned Cu ($Cu\ 2p_{3/2}$ = 932.62 eV and $Cu\ 3p_{3/2}$ = 75.1 eV) and Au foils ($Au\ 4f_{7/2}$ = 83.96 eV). ~~Peaks were~~ Charge referenced to the CH_x band in the carbon 1s peak was assigned at 284.8 eV. Measurements were made at a takeoff angle of 45° with respect to the sample surface plane. This resulted in a typical sampling depth of 3−6 nm, which contains 95% of the signal originated from this depth. Quantification was done using instrumental relative sensitivity factors that account for the X-ray cross-section and inelastic mean free path of the electrons.

The contact angle analyzer (DSA4; Kruss) with a high-resolution camera was used to evaluate the surface hydrophilicity of the PCL-aligned fibers and the untreated glass slide.

2.4. Mechanical Testing. The fiber samples produced by both touch-spinning and electrospinning techniques were collected on 10 mm × 30 mm rectangular frames. An electromechanical universal test system (MTS Criterion Model 42) was used for testing the nanofibers at a crosshead speed of 2 mm min^{-1}. All prepared PCL fibers were further cut into 10 mm × 18 mm rectangular sheets, and these were locked on each end by the mechanical grips. Five parallel measurements were conducted for each fiber sample. The Young's modulus was calculated by dividing the force on the fiber by the effective cross-sectional area of the fiber, A. The effective cross-sectional area, A, was determined as

$$A = \frac{W}{(L \times PD)} \quad (1)$$

where W is the weight of the fiber, L is the length of the fiber, and PD is the density of PCL (1.145 g cm^{-3}).

2.5. Culture of NE-4C Cells on Nanofibrous Scaffolds. NSCs were maintained in Eagle's minimum essential medium (ATCC) supplemented with 5% fetal bovine serum (FBS, ATCC) and 1% L-glutamine (Sigma-Aldrich). The cells were cultured in the 150 cm^2 flask in an incubator under a humidified atmosphere with 5% CO_2 at 37 °C. The culture medium was changed every 2 days. Subconfluent cultures were regularly passaged by trypsinization using 0.05% trypsin (Sigma). Prior to cell seeding, the nanofibrous scaffolds were sterilized by 70% ethanol for 30 min, rinsed and immersed in PBS overnight, and prewetted in poly-l-lysine at 37 °C. NSCs were seeded at a density of 30 000 cells per scaffold for a 4 h cell attachment. NSC proliferation over the 1, 4, and 7 days was performed as well. After each prescribed time interval, scaffolds were transferred to new well-plates for viability tests. Cell viability was measured using the Cell Counting Kit-8 from Dojindo Molecular Technologies after a 2 h incubation at 37 °C. Subsequently, a Thermo Scientific Multiskan GO Microplate Spectrophotometer was used to measure the absorbance at the wavelength of 450 nm. All cell viability experiments were repeated three times with three replicates per group, totaling 9 samples per group.

2.6. Preparation and Fluorescent Immunocytochemical Staining of NE-4C Cells. NE-4C NSCs were cultured in the respective standard medium with 10^{-6} M RA for 96 h to induce neuronal differentiation. After an initial 96 h RA treatment, the cells were maintained in the standard medium without RA for an additional 4 days. All PCL fibers were prewetted with control media overnight to enhance the cellular attachment. After each culture period, samples were gently rinsed with PBS and fixed with 10% formalin for 12 min at room temperature. The samples were

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further permeabilized with 0.2% Triton X-100 in PBS for 5 min. Diluted mouse anti-TUJ1 (1:1000; Covance) primary antibody was gently added in scaffolds, and it was incubated at 4 °C overnight. After three wash cycles with PBS to remove any unbounded primary antibody, samples were incubated in secondary antibody with Alexa Fluor 594 goat anti-mouse (Life Technologies) at room temperature. The cell nuclei were stained by 4′-6-diamidino-2-phenylindole dihydrochloride (DAPI, 10 μg mL^{-1}) (Life technologies). Laser scanning confocal microscope (LSCM 710, Zeiss) controlled via the Zen software was used to detect and monitor the high-resolution fluorescent images of the 3D growth of neural cells and neurite extension of the cells.

2.7. Quantification of Neurite Angle and Length. ImageJ[32] (ImageJ; National Institutes of Health, Bethesda, MD) and the NeuriteTracer[33] plug-in were utilized to determine the angle of neurite outgrowth relative to the direction of fiber alignment. Neurite expression levels detected from TUJ1 staining were also quantified with the ImageJ software. Linear color level adjustments were applied identically to all samples in the Zeiss Zen software. Four areas were randomly selected on each sample (6 samples/group) for analysis using a 10× objective. A total of 37 to 441 cells/group were observed with nuclear staining and TUJ1 positive cells.

2.8. Statistical Analysis. All quantitative results are expressed as the mean ± standard error of the mean. Statistical significance between the groups was tested by the student's *t*-test. Statistical difference was considered at $p < 0.05$.

3. RESULTS

3.1. Characterization of the Prepared PCL Nanofibers. FEI Teneo field emission scanning electron microscopy was utilized to observe the morphology of the PCL nanofibers fabricated by electrospinning and touch-spinning. Figure 2a represents the aligned PCL nanofibers prepared by electrospinning and touch-spinning methods. The aligned PCL fibers were successfully fabricated from an 8 wt % PCL solution. The Fiji[34] software was used to measure the diameter of the nanofibers. The average diameters of 50 electrospun and touch-spun nanofibers were 481 ± 80 and 431 ± 30 nm, respectively.

The crystal structures of the PCL nanofibers prepared by two separate spinning techniques were confirmed by XRD (Figure 2b–d). Positions of two picks for both electrospun and touch-spun PCL nanofibers were identical at 2θ = 21.5 and 23.9°, respectively, suggesting no change in the crystal structure by fabrication techniques. However, the intensities of diffraction peaks for touch-spun PCL fiber were higher than those of the electrospun PCL fiber, indicating a higher degree of crystallinity in the touch-spun fiber compared to that of the electrospun fiber (Figure 2b). Figure 2c,d shows the fitted (110) diffraction peaks with FWHM in degree, and Figure 2e shows the crystallite size of PCL nanofibers. The crystallite size of the touch-spun fibers was found to be 311.5 Å, which was smaller than that of the electrospun fibers (around 350 Å).

XPS result is summarized in Table 1. Two different concentrations (0.1 and 5%) of BSA incorporated in PCL fibers were analyzed to detect a difference in the surface atomic composition. Species detected on all PCL fiber surfaces included: CH_x, C–N, C–O and (N,O)–C=O consistent with mixtures of PCL and BSA. Particularly, the N levels, which represent the presence of BSA, were markedly higher on the two electrospun samples (3.2 and 3.2) compared to the two touch-spun samples (1.4 and 0.3). Interestingly, a higher concentration of BSA encapsulation in PCL fibers did not result in a higher concentration of N on both electrospun and touch-spun samples.

An aligned extracellular matrix is a common feature of the biological tissue, such as the tracts of aligned cells present within the nervous system.[35] The development of an aligned ECM architecture of the native nerve tissue with the appropriate restoration of mechanical properties is crucial in clinical repair.[35,36] In this regard, both electrospinning and touch-spinning are suitable fabrication techniques as they can produce well-aligned fibrous scaffolds, which is important for applications involving mechanically anisotropic soft tissues. Specifically, we employed a uniaxial mechanical testing system to measure the mechanical properties of the PCL nanofibers (Figure 3). It should be noted that Young's modulus of the

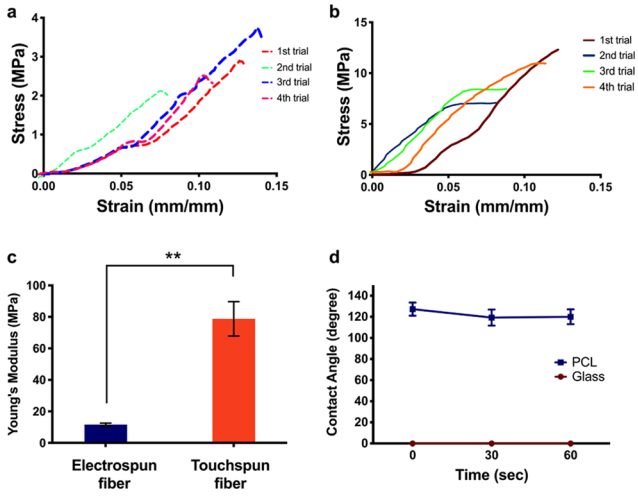


Figure 3. Mechanical and surface properties of nanofibers. The stress–strain curve of (a) electrospun and (b) touch-spun PCL nanofibers. (c) Young's modulus values for PCL nanofibers. Data are the mean ± standard error of the mean. n = 4; $^{**}p < 0.01$ when compared to the electrospun fiber. (d) The contact angle of PCL touch-spun fibers when compared to glass. Data are the mean ± standard error of the mean; n = 4.

touch-spun fibers was significantly higher than that of electrospun fibers, which is in agreement with higher crystallinity of touch-spun fibers (Figure 3c). Specifically, Young's modulus of PCL touch-spun fibers was 6-fold greater than that of electrospun fibers. Obviously, the fiber drawing conditions in the touch-spinning process are favorable for polymer chain alignment by shear forces.

The static contact angles of a water droplet (5.0 μL) on the untreated glass slide and the PCL touch-spun fiber were 0 and 127.3 ± 6.2°, respectively. The contact angle slightly decreased on PCL fiber substrates 30 and 60 s after the droplet fell on the surface. The contact angle of the PCL touch-spun fiber at 60 s was 120.1 ± 7.0°, whereas the contact angle on the glass slide remained at 0°, demonstrating its relative hydrophobic property (Figure 3d).

3.2. Biocompatibility Analysis of the PCL Nanofibers. The 4 h attachment study showed that the NSC adhesion increased by 50% on both electrospun and touch-spun fibers compared to the glass substrate (Figure 4a); however, there was no statistical difference between the electrospun and the touch-spun fibers. For the subsequent studies, we incorporated BSA into each PCL fiber to investigate its effect on NSC growth. A CCK-8 assay was performed to evaluate the viability of NSCs on the electrospun and touch-spun PCL fibers. The CCK-8 reagent is reduced by dehydrogenase in cells to give

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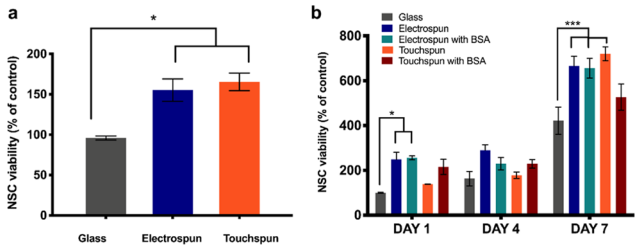


Figure 4. NSC growth on PCL nanofibers. (a) NSC adhesion on various substrates after 4 h of culture. Data are the mean ± standard error of the mean; $n = 9$, $^{*}p < 0.05$ when compared to glass. (b) Enhanced NSC proliferation on glass, electrospun fibers, electrospun fibers with BSA, touch-spun fibers, touch-spun fibers with BSA after 7 days of culture. Data are the mean ± standard error of the mean, $n = 9$; $^{*}p < 0.05$ when compared to glass on day 1, $^{***}p < 0.001$ when compared to glass on day 7.

orange-colored formazan, providing information about cell growth and metabolic activity of the cells. Figure 4b shows the 7 day proliferation of the NSCs on the resultant scaffolds. The NSCs cultured on both electrospun and touch-spun PCL fibers increased by 58 and 71%, respectively, compared to the glass control group on day 7. Despite the low initial cell numbers found on the touch-spun fibers on day 1, NSCs proliferate significantly after day 4 with a doubling time of 1.5 days. In contrast, NSCs cultured on electrospun fibers had a doubling time of 2.6 days. However, there was no statistically significant difference between electrospun fibers and touch-spun fibers on day 7. Furthermore, the incorporation of BSA did not affect NSC proliferation between electrospun fibers and touch-spun fibers throughout the 7 days of cell culture.

3.3. Early Differentiation of NSCs on the PCL Nanofibers. Confocal microscopy images showed early stages of the neural differentiation stained by TUJ1 (Figures 5, 6, 7). We first observed enhanced neural alignment on the touch-spun PCL nanofibers at day 8 of early differentiation (Figure 5a). Figure 5c shows that the touch-spun fibers provide significantly enhanced neurite alignment on the fibers. 14% of randomly measured neurite segments extended directly along the direction of fibers. In addition, 41% of neurite segments on touch-spun fibers showed a high degree of alignment, within 5° of the fiber direction. In contrast, electrospun fibrous scaffolds showed a noticeably lower level of neurite alignment (Figure 5b), where 52% of all measured neurite segments diverged more than 20° from the aligned fibers.

We also incorporated BSA into each kind of PCL fibers to observe its effect on NSC differentiation. After 4 days of RA induction, small amounts of TUJ1 immunostaining appeared on all constructs. The NSCs cultured on the glass substrate maintained a spherical cluster, whereas NSCs cultured on the aligned fibers elongated along the fibers. Figure 6a shows the early neural differentiation of NE-4C cells on various PCL nanofibers. However, by Day 4, it was difficult to see the orientation of NSCs along the fibers. Figure 6b represents TUJ1 relative expression on the PCL fibers vs. the glass slide. As shown in Figure 6a, higher TUJ1 expression levels were more evident on the PCL fibers than on the glass slide. NSCs cultured on the glass slide expressed about 1.6 TUJ1 levels compared to ∼15 on the PCL fibers. There was a significant difference between the various PCL fibers. The ratio of TUJ1 and DAPI was also consistent with the relative expression levels of TUJ1 (Figure 6c). The percentage of TUJ1 positive

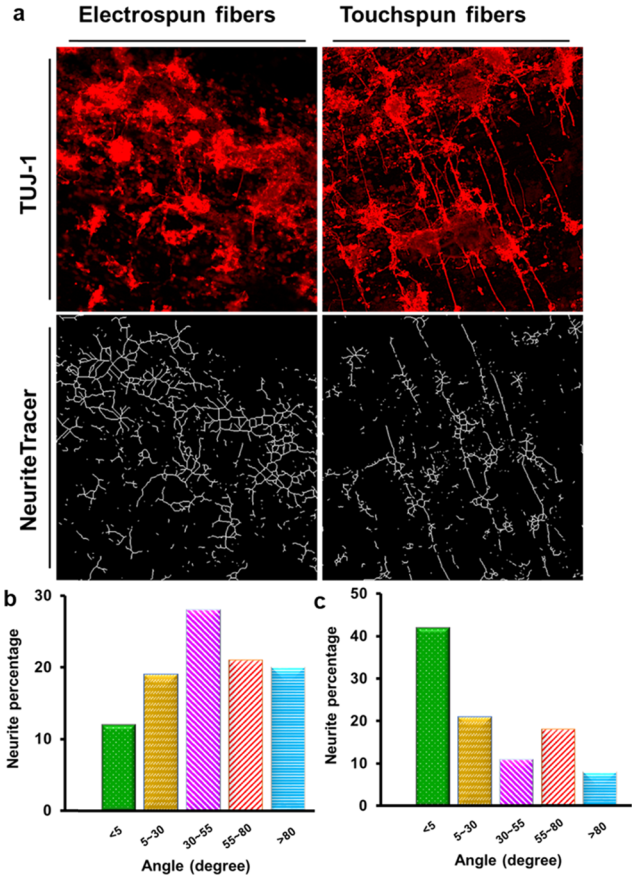


Figure 5. Confocal microscopy images of NSC attachment and spreading on PCL nanofibers. (a) Cells were stained with TUJ1 (red) for electrospun fibers and for touch-spun fibers. ImageJ with NeuriteTracer Plug-in was used to trace the associated neurite outgrowth. The neurite angle distribution on the corresponding (b) electrospun PCL fibers and (c) touch-spun fibers. The angles between PCL nanofibers and neurites have been shown as <5, 5−30°, 30−55°, 55−80°, >80°. $N = 6$.

cells was greatly increased on all PCL nanofibers compared to that of the glass slide (Figure 6c). However, it should be noted that the incorporation of BSA in touch-spun fibers did not appear to change the TUJ1 expression levels by day 4.

By the 8th day of neural differentiation, the clusters of TUJ1 positive cells were apparent throughout the substrate (Figure 7a). TUJ1 positive cells on the glass slide tended to form a large aggregate, and these remained within the boundary of DAPI positive cells. TUJ1 positive cells on the touch-spun PCL fibers tended to form a smaller aggregate along the fiber compared to that of electrospun PCL fibers (Figure 7b,c). More importantly, longer neurite extension was more noticeable on the touch-spun PCL fibers connecting between the aggregates (Figure 7c). When comparing the TUJ1 relative values, NSCs cultured on the touch-spun fibers with BSA had a significantly higher value compared to any other group (Figure 7d). Consistently, the value of TUJ1 expression per DAPI within cells on the touch-spun fibers with or without BSA significantly increased compared to glass and electrospun fibers by day 8 (Figure 7e).

4. DISCUSSION

4.1. Touch-Spinning is a Viable Alternative to Electrospinning for Creating a Neural Scaffold. In this

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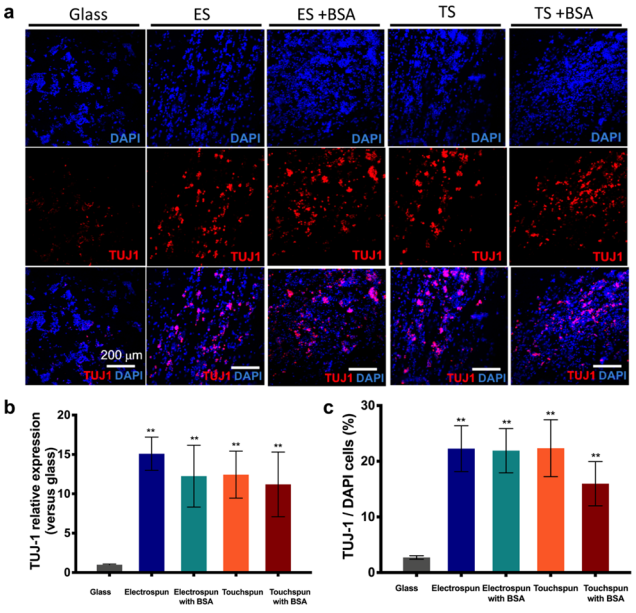


Figure 6. Early differentiation of NSC at Day 4. (a) Immunocytochemical staining of TUJ1 (red) and DAPI (blue) of NSC on various substrates after 4 days of differentiation. (b) TUJ1 relative expression of seeded NSC on various substrates. (c) Quantification of the percentage of TUJ1 cells per nuclei on day 4. Data are the mean ± standard error of the mean, $n = 6$; $^{**}p < 0.01$ when compared to glass. Scale bar = 200 μm. es = electrospun fibers; TS = touch-spun fibers.

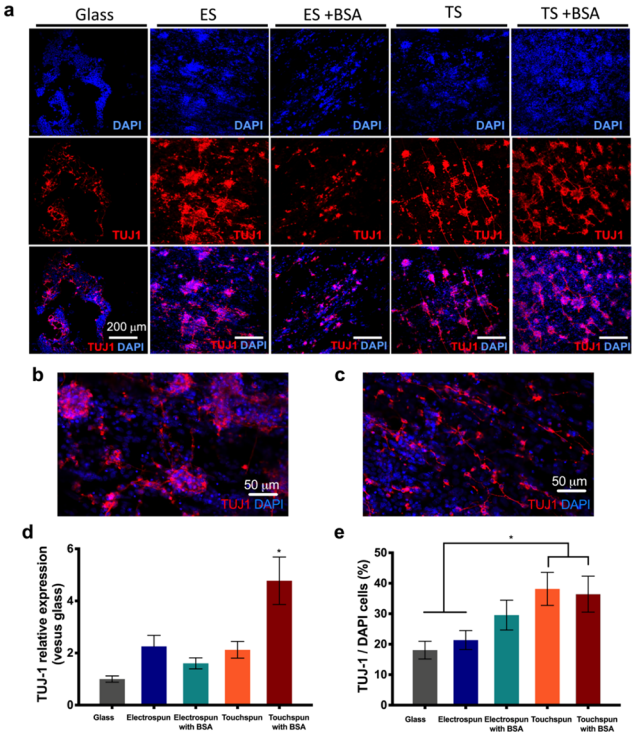


Figure 7. Early differentiation of NSC on day 8. (a) Immunocytochemical staining of TUJ1 (red) and DAPI (blue) on various substrates after 8 days of differentiation. (b) and (c) High-magnification images of TUJ1 and DAPI on electrospun and touch-spun fibers, respectively. (d) TUJ1 relative expression of seeded NSC on various substrates. (e) Quantification of the percentage of TUJ1 cells per nuclei on day 8. Data are the mean ± standard error of the mean, $n = 6$; $^{*}p < 0.05$. Scale bar = 200 μm. es = electrospun fibers; TS = touch-spun fibers.

study, we generated PCL nanofibers using two different fabrication methods, including the conventional electrospinning method and a novel touch-spinning[31] technique. These two methods were used to fabricate nanofibers of the identical chemical composition and similar geometrical shape but with different tensile strength affected by differences in the morphology of the nanofibers. We may speculate that a superior nerve scaffold should have mechanical strength similar to that of native nerve tissues to withstand manual manipulation and in vivo physiological loading during nerve regeneration, as well as to retain its structure after implantation.[37] Liu et al. showed recently that a human sciatic nerve in the longitudinal direction exhibits Young's modulus in a range of 40.96 ± 2.59 MPa.[38] The fabricated touch-spun and electrospun fibers showed Young's moduli of 79 ± 11 and 12 ± 0.9 MPa, respectively, indicating the favorable tensile properties of touch-spun fibers.

The higher degree of crystallinity of the touch-spun fibers is mainly due to a higher shear force developed during the drawing of the fibers at the stage of their solidification upon evaporation of the solvent. The effects of all factors during fiber formation, including the speed of solvent evaporation and changes of elastic properties of the liquid thread leveraging the pulling force, are highly nonlinear. Obviously, similar drawing conditions could be approached using different fiber drawing methods. However, it turned out that the touch-spinning technique is quite flexible to adjust and tune the fiber formation conditions. Therefore, it was possible to reveal a range of fiber spinning parameters providing polymer chain stretching and alignment, which enhanced the degree of crystallinity and elastic modulus of the touch-spun nanofibers. It was also found that the crystallite size of the touch-spun PCL fibers (310 Å) is smaller than that of the electrospun fibers (around 350 Å) indicating a better alignment in the first case resulting in an increased concentration of crystallization nuclei.[39]

4.2. Directed Neurite Outgrowth. Figure 5 shows the confocal micrograph and neurite traced images of TUJ1 expression of NSCs cultured on the highly aligned PCL touch-spun and electrospun nanofibers. The result exhibits a higher order of alignment of NSCs on the touch-spun fibers compared to the electrospun fibers. Furthermore, most of the differentiated NSCs exhibited bipolar morphology with two extended neurites on the touch-spun fibers, whereas NSCs cultured on the aligned electrospun fibers expressed multipolar processes. These results suggest that the aligned neurite outgrowth of NE-4C cells might be mainly guided by the chemical or physical cues of PCL touch-spun nanofibers since there was no obvious oriented neural growth provided by the aligned electrospun fibers.

As we discussed previously, the nanofibers produced by the touch-spinning method revealed different structural characteristics as compared with electrospun fibers. Among those altered features, high crystallinity properties of the touch-spun fibers may play an important role in governing NSC growth, spreading, and alignment during the early stage of the neural differentiation.[24] The mechanism of this cell response to crystallization-induced surface properties is unclear yet. But it is well documented that polymers with higher crystallinity affect the cytoskeletal organization and subsequent cell phenotype.[23,40−42] Some studies found that highly crystalline or stiffer substrates were found to improve cell spreading, proliferation, focal adhesion maturation, and differentiation of

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various stem cells.[23,24] Cui et al. showed that crystallinity and rigidity of the polymers played significant roles in determining cell response when chemical composition, surface roughness, and hydrophobicity of PCL/PGA polymers remained constant.[24] In that study, highly crystalline and rigid PCL and PGA substrates significantly increased fibroblast growth compared to the amorphous polymer surface. However, it should be noted that other types of cells, such as osteoblasts, were found to favor less crystalline substrate for growth and spreading.[24]

The molecular mechanisms involving mechanotransduction in neuronal cell development are still largely unexplored. Physical stimuli signals such as stiffness and crystallinity in the neural differentiation of stem cells lead to a cascade of signaling events including Rho GTPases signaling (i.e., RhoA, Rac1, and Cdc42) with cytoskeleton regulators, such as Yes-associated Protein (YAP).[43] Multiple studies have identified the Rho GTPase-mediated cytoskeletal dynamics and the transcriptional co-activator YAP as essential players in stiffness-instructed neuronal differentiation.[44−46] On a stiff substrate, suppression of RhoA and activation of Rac1 and Cdc42 promote neurogenesis in the NSC.[46] However, it should be noted that previous investigations have been focused on the effect of soft materials with modulus less than 5 MPa. No studies have investigated the influence of high crystalline nanofiber with a modulus range of 1−100 MPa on the differentiation of NSCs. Among several possible explanations, we hypothesize that due to the high elastic modulus coupled with a high degree of crystallinity of touch-spun fibers, actin filaments reorganize and activate the Rac1 and Cdc42. The activated Rho family of GTPases then phosphorylates YAP to encourage alignment and neurogenesis of the NE-4C cells. Future studies will be conducted to elucidate the mechanotransduction mechanism activated in multiple stages of NSC differentiation by the aligned touch-spun PCL nanofibers with controlled modulus and crystallinity.

4.3. BSA-Incorporated Touch-Spun Enhanced Early Differentiation of NSCs. Serum albumin is the most abundant protein component of blood plasma, and it has shown to play a crucial role in tissue development.[47,48] BSA has been widely used in cell culture experiments, and it has been found to be present during the early stages of brain development. Some studies proposed the importance of albumin for its role in NSC differentiation.[21,49] Furthermore, it is well-documented that BSA induces calcium waves and the proliferative rate of astrocyte. Tabernero et al. showed that BSA can strongly increase the flux of glucose and lactate through the pyruvate dehydrogenase-catalyzed reaction in the primary cultured astrocytes.[21] In the present study, 0.01 w/v % BSA was blended into the PCL solution, and the fibers were fabricated by electrospinning and touch-spinning methods. We examined how incorporated BSA affects early neuronal differentiation of NSCs by observing the ß-tubulin class III expression level on the resultant fibers.

When observing NSCs on the touch-spun PCL nanofibers with incorporated BSA, it was discovered that the ß-tubulin class III expression level per nuclei was enhanced compared to those on the electrospun PCL nanofibers after 8 days of culture (Figure 7e). When looking at ß-tubulin class III expression levels alone, NSCs cultured on the touch-spun nanofibers with BSA had significantly higher expression values than any other group. This is indicative that NSCs cultured on the touch-spun fibers with BSA were more likely to be differentiated into neuronal lineages as compared to those cultured on the electrospun fibers with or without BSA. There are several possible mechanisms to explain the role of BSA incorporated in the touch-spun fibers in an early neuronal differentiation compared to the electrospun fibers. An increase in the early differentiation of NSCs in the touch-spun fibers with BSA may be partially due to the conformational stability of the protein. The unfolding of proteins results in a loss of enzymatic activity and, in many cases, formation of amyloid fibrils.[50] Some studies have shown that even small electric fields induce changes in BSA conformation.[50,51] Electrospinning involves a high voltage, which may diminish the bioactivity of incorporated growth factors by affecting conformational stability as well as enzyme activity.[52] A study by Ji et al. showed that after electrospinning, the fiber samples showed a significant decrease in enzyme activity of alkaline phosphatase (ALP) compared to freshly dissolved ALP. The secondary structure analysis further confirmed that the incorporated proteins in the electrospun fibers underwent conformational changes, suggesting that high voltage may be harmful to the loaded biomolecules.[53] In contrast to the conventional electrospinning, the touch-spinning method involves no voltage to produce highly aligned polymeric nanofibers, which may be helpful in preserving the activity and conformation of loaded growth factors. In future studies, we will thoroughly examine conformational changes of the secondary structures and release kinetics of incorporated bioactive factors in the fabricated PCL nanofibers.

5. CONCLUSIONS

Touch-spun PCL nanofibers exhibited a higher degree of crystallinity and elastic modulus over the fibers obtained by conventional electrospinning. Incorporated BSA within the touch-spun fibers increased the expression of neuron-specific class III β-tubulin compared to other scaffolds by day 8. More importantly, the aligned PCL touch-spun nanofibers promoted aligned neurite outgrowth of NSCs, suggesting its potential utility in treating conditions ranging from spinal cord injury to peripheral nerve lesions.

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Author Contributions

L.G.Z. and S.M. directed the project and initiated the main idea for this study. S.-J.L. performed the cell culture experiments, analyzed the proliferation and differentiation of the NSC cells, and measured the modulus of the nanofibers. D.A. fabricated the touch-spun and electrospun nanofibrous at various spinning rates. R.T. analyzed the degree of crystallinity according to XRD results. All authors contributed to the analysis of the results and commented on the manuscript.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

PCL, polycaprolactone
NSCs, neural stem cells
XPS, X-ray photoelectron spectroscopy
XRD, X-ray diffraction
BSA, bovine serum albumin
FBS, fetal bovine serum
PBS, phosphate-buffered saline
RA, retinoic acid

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